

Future Planet - How the present can influence the future

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Figure 1: Player in present (left) can execute actions that impact the environment of the player in the future (right)

ABSTRACT

“Future Planet - How the present can influence the future” is an interactive and collaborative virtual environment designed to raise awareness of environmental issues and promote sustainable choices. The project addresses critical issues such as deforestation, air pollution and clean energy and challenges users to solve them in order to create a sustainable future. Users make good or bad decisions to learn about the impact on the environment - for example, planting appropriate trees combats deforestation, while poor building decisions increase pollution. By completing tasks, such as discovering hidden seeds or collecting research papers, players unlock items that they can use to develop their future planet. Cooperation is important: one player in the present makes decisions that affect the future, while another in the future observes the effects and provides assistance. Interaction features include a gesture-based 3D user interface with a miniature earth for teleportation and unique tools for navigation and item management. By combining education, gamification and immersive elements, the project encourages users to explore sustainable behaviors and work together to create a greener, more sustainable planet.

Index Terms: Sustainability, 3DUI, Collaboration, VR.

1 INTRODUCTION

As concerns about climate issues continue to grow, the video game industry has been exploring solutions to raise public awareness

about the impact of individual lifestyles on the environment. The development of virtual reality technologies enables new immersive experiences and innovative ways of imparting knowledge.

2 RELATED WORK

Previous research has shown the benefits of virtual reality (VR) in educational settings, as VR can strongly engage users in knowledge acquisition, leading to better learning outcomes [3]. Adding a collaborative dimension to immersive learning allows users to become even more involved as they share common experiences and difficulties in accomplishing tasks. Previous studies have particularly highlighted the significant relationship between social presence and cognitive engagement [2]. In the context of environmental issues and climate change, several projects have highlighted the potential of VR to raise public awareness in a playful yet impactful way. The Stanford Ocean Acidification Experience is a large-scale educational VR application that allows users to experience the damage of ocean acidification due to high CO₂ emissions [4]. On a smaller scale, other applications propose, for example, to visualize the user’s carbon footprint in VR to combine immersive technologies with location-based environments to study climate change [5], each time with effective results. Studies show that the quality of a virtual environment significantly influences user perception, as shown in “Presentational Validity of Landscape Visualizations” [1]. We enhanced the visual elements of our project to motivate users to protect the world they are exploring and make the exploration engaging with hidden elements.

3 CONCEPT

The general concept of the virtual environment is that two players have to solve environmental problems in three different areas, each focusing on one main issue: Deforestation, Clean Energy and Water Pollution. The two players take on different roles in two different timelines - one in the present and one in the future. Their goal is to work together and make the right decisions in the present to save the future. Cooperation is encouraged by solving various puzzles and making important decisions.

The decision-making process aims to make players aware that the choices we make today directly affect the future, whether pos-

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itively or negatively. For example, if a player plants a seed in the present, a tree will grow in the future and improve environmental conditions, or if a different method of energy production is chosen, air quality will improve in the future. A miniature earth with teleportation points is available for navigation in the various areas. Interaction with the environment is made possible by objects in the world and an inventory system containing various items. These functions can be controlled with gestures and gesture sequences, ensuring an intuitive and user-friendly experience in the virtual environment.

4 SYSTEM AND 3D USER-INTERFACE DESIGN

The virtual environment was developed with Unity and is designed for use with Meta Quest headsets, as they have hand tracking capabilities.

Players can navigate between levels using a small planet that they can summon by opening their left hand. This planet has several red markers on its surface, as shown in figure 2.



Figure 2: The miniature Earth can be used for teleportation to different areas.

By touching these markers with their index finger, players can teleport between levels and begin their exploration. To reach all available teleportation points, it is necessary to rotate the globe. For this, we utilised Meta Quest 3's hand tracking features, which allow the user to grab predefined virtual objects by performing a pinching motion when interacting with them, as shown in Figure 3.

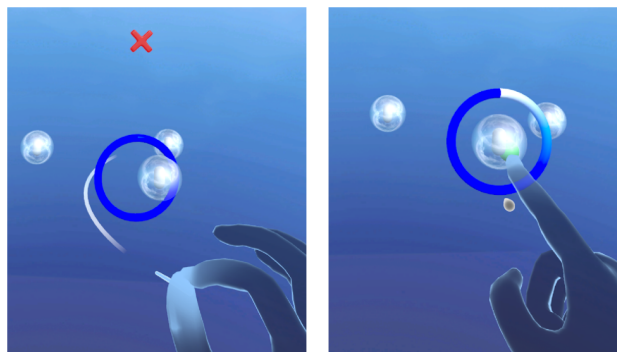


Figure 3: The inventory can be moved for the selection (left) and tapping on an item selects it (right).

These interactions allow the user to interact directly with the virtual environment, which improves immersion in the application. Our inventory system also relies on this hand tracking system. For example, the player must turn his right hand towards the sky, similar to the mechanics of the small planet, and interact with it through

touch and pinch gestures. This system enables direct interaction between the player and the environment, meaning that no peripheral devices other than the VR headset are required. Simplifying the controls allows players to focus on their surroundings and work together to solve the tasks presented to them.

The figure 4 illustrates the visual differences between the present and future environments. The future appears significantly gloomier in contrast to the present. The concept incorporates a transformation mechanism, where successfully completing tasks in each environment results in a better future, visually represented by a healthier and more vibrant map.

Although the two players are transported to two different eras, they can concentrate on the tasks set for them thanks to the same simple interaction options with their surroundings.



Figure 4: Comparison between the present and future city environment.

5 RESULTS AND NOVELTY

The described techniques and design decisions resulted in a collaborative and innovative concept for both players. By integrating collaboration between players in the present and future with a dynamic decision-making process, combined with a 3D user interface and interactive environments, a unique gaming experience with a strong environmental focus was developed. The immediate feedback that links present actions to future outcomes reinforces the concept of environmental responsibility in an engaging and impactful way. Through gesture-based interaction and an innovative concept, the two players experience a completely new form of gameplay that raises their awareness of the importance of today's choices. The overarching goal of sensitising players to environmental choices while providing a unique and intuitive gaming experience has been achieved.

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